Ninapro DB3 sEMG: Exploratory Data Analysis and GNN Related Work

1. Summary

This report analyses Ninapro DB3, a surface EMG (sEMG) dataset of hand gestures recorded from transradial amputees, as preparation for graph-neural-network (GNN) gesture classification. Raw signals were cut into 150 ms windows with 50% overlap, and six time-domain features (MAV, RMS, VAR, WL, ZC, SSC) were computed for each of 12 channels, giving 72 features per window.

The main findings are:

- The dataset is large and has no missing values, but it is strongly imbalanced because the rest class dominates.
- The amplitude features (MAV, RMS, WL) are right-skewed and heavily correlated within a channel, so about half of the 72 features are redundant.
- Subject identity is likely to be a strong confounder, so evaluation must be subject-wise.
- The original ZC/SSC extraction was faulty and has been fixed.

2. Dataset and pipeline

Dataset	Ninapro DB3 (Ninapro project, Atzori et al., 2014)
Subjects	11 transradial amputees
Item	Value
Channels/rate	12 sEMG channels, 2 kHz
Files	33 expected, 32 loaded (first run); `S3_E1_A1.mat` is corrupted
Raw signal points	53,103,168 (first run)
Window	300 samples (150 ms), step 150 (50% overlap), uniform-label windows only
Features	MAV, RMS, VAR, WL, ZC, SSC × 12 channels = 72
Units	Signal converted from V to mV; ZC/SSC threshold = 0.1 × channel std

Name	Ninapro DB3
Source	Ninapro project (ninapro.hevs.ch)
Area	Biomedical signals, sEMG hand-gesture recognition
Samples (windows)	341,970 (first run) / [re-run]
Classes	50 (rest plus 49 gestures)
Features	72 numeric time-domain features
Subjects	11
Balance	37.5:1 overall (first run); active-only ratio [re-run]
Missing values	0 (first run)
Attribute	Value
Duplicate rows	0 (first run); [re-run]

4. Track B: Statistics

The notebook computes the mean, median, standard deviation, minimum, maximum, and quartiles, skewness, excess kurtosis, and IQR-based outlier counts for all 72 features, and saves them to feature_statistics.csv. The first run showed:

- MAV values of order 1e-5 (volts), with maxima up to about 4e-3, so the distributions have very long right tails.
- A mean MAV that is far above its median in most channels, which confirms strong positive skew.
- An average outlier rate of 8.17% per feature (**first run**). This is understated because the all-zero ZC and SSC columns counted as 0% outliers. Use the **[re-run]** value.

[re-run] Insert the per-type median table (MAV, RMS, VAR, WL, ZC, SSC) from cell B.

5. Track C: Data quality

Check	Result	Effect on modelling
Missing values	0 (**first run**)	No imputation needed
Duplicate rows	0 (**first run**)	None; confirm after re-run
Near-constant features	35 flagged (**first run**), invalid	Caused by the threshold bug and a scale-dependent variance test; **[re-run]**
Highly correlated pairs ($\|\rho\| \geq 0.95$)	20 (**first run**), Pearson	**[re-run]** with Spearman; MAV↔RMS was 0.977–0.997 per channel
Outliers	8.17% average (**first run**)	Real muscle bursts; prefer robust or quantile scaling over clipping
Skewness	High for MAV, RMS, WL, VAR	Log or quantile-transform before distance-based methods

6. Track D: Class balance

- Rest (label 0) is the majority class: 128,864 windows, 37.68% (**first run**).
- The smallest class is label 17: 3,434 windows, 1.00% (**first run**).
- Overall imbalance is 37.5:1; the active-gesture-only ratio is **[re-run]**.
- A classifier that always predicts rest scores 37.7% accuracy but a near-zero macro-F1, which is why accuracy is misleading here.

7. Track E: Feature distributions

We plotted histograms with KDE, boxplots, and violin plots for MAV, RMS, WL, and ZC on channel 1 for gestures 1, 2, 5, 8, and 12. Amplitude features are plotted on a log10 scale. Class separation is quantified with η^2 (share of feature variance explained by gesture class), both raw and after per-subject z-scoring.

[re-run] Insert the η^2 values printed by cell E. The reading rule is: below 0.06 is weak, 0.06–0.14 moderate, above 0.14 strong. The comparison of raw against per-subject values shows how much of the apparent class signal is actually subject differences.

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8. Track F: Correlation and redundancy

Spearman correlation was used because the features are skewed. The first run found strong MAV↔RMS redundancy within each channel (ρ between 0.977 and 0.997). The corrected notebook outputs an explicit drop list (the later-listed feature of every correlated pair, plus near-constant features) and a reduced feature set.

[re-run] Insert the number of redundant pairs, the drop list length, and the reduced feature count.

9. Track G: 2D projections (PCA, t-SNE, UMAP)

Eight gestures (1–8) were sampled at equal counts, quantile-transformed to normal, and projected with PCA, t-SNE, and UMAP. Separation is measured with silhouette scores for gesture labels and for subject labels.

The first run's PCA showed heavy overlap. The earlier claim that t-SNE "proves" high accuracy was unsupported and has been removed; a clustering picture shows structure, not classifier performance.

[re-run] Insert gesture and subject silhouette for PCA, t-SNE, and UMAP. If the subject silhouette is higher than the gesture silhouette, subject identity dominates the feature space.

10. Track H: Interactive plots

Six Plotly figures are provided, each with a title and a one-line reading: class balance bars, an RMS violin by gesture, a correlation heatmap, a 2D UMAP, a 3D PCA, and a 3D UMAP, all with subject shown on hover. In the first run, the first three principal components kept 64.4% of the variance (this included constant columns). **[re-run]** Use the corrected value.